

# SAKSHAM GUPTA

1-C, CK-5 Chanakya Rail Enclave, Chanakyapuri, New Delhi

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## EDUCATION

**Manipal Institute of Technology, Manipal**

*Manipal, Karnataka • May 2027*

**B.Tech in Information Technology**

CGPA: 9.16/10.0

**Delhi Public School R.K. Puram**

*New Delhi • 2023*

Class 12 – CBSE (90.2%)

Class 10 – CBSE (93.5%)

## TECHNICAL SKILLS

**Programming Languages:** Java, Python, JavaScript, C++, C

**Computer Science Fundamentals:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Time & Space Complexity Analysis

**Frontend:** React.js, HTML, CSS, JavaScript, Socket.io, Bootstrap

**Backend:** Java (OOP), Flask, Node.js, Express, RESTful API Development

**Software Engineering:** SDLC, Agile, Unit Testing, Integration Testing, UML

**Databases & DevOps:** MySQL, PostgreSQL, MongoDB, Git, GitHub Actions (CI/CD), Docker

**Systems & Security:** Linux, Bash, Multithreading, Concurrency Control, AES-256 Encryption, Digital Signatures

## EXPERIENCE

**Software Development Intern**

*June – July, 2025*

**Indian Railway Catering and Tourism Corporation (IRCTC)**

- Analyzed enterprise booking system architecture handling 2M+ daily transactions, identifying data flow and performance bottlenecks
- Optimized SQL query execution plans and indexing strategies, reducing response latency by 25% in high-volume transaction environment
- Evaluated scalable system design patterns including microservices architecture for high-traffic applications
- Developed Python-based load testing scripts to simulate concurrent transactions and measure system throughput

## PROJECTS

**SafePay | High-Throughput Transaction Fraud Detection System | [GitHub](#) • [View Live](#)**

*Python, Flask, MongoDB, Scikit-learn, Cryptography*

- Developed a secure full-stack payment system with Flask REST APIs and integrated ML-based fraud detection (85%+ accuracy)
- Implemented AES-256 encryption, digital signatures, and concurrency-safe balance updates to ensure transactional integrity
- Architected a modular backend system and implemented unit tests for critical transaction workflows

**Fileshare | Real-Time Anonymous File & Chat Sharing | [GitHub](#) • [View Live](#)**

*Node.js, Express, Socket.io, React, MongoDB, HTML, CSS, JavaScript*

- Built a full-stack real-time application using React frontend and Node.js/Express backend with Socket.io for live communication
- Implemented REST APIs and room-based session logic with automatic expiry for privacy and scalability
- Deployed production-ready system to cloud with CI/CD integration and structured error handling

**Starfinder | Multi-Class Stellar Classification System | [GitHub](#) • [View Live](#)**

*Python, Flask, Scikit-learn, Pandas, NumPy, JavaScript*

- Developed multi-class stellar classification using Random Forest, achieving 93% accuracy across 6 star types
- Built input validation to handle out-of-distribution data and maintain prediction reliability
- Deployed a production ML system with Flask REST API, automated model persistence, and data analysis pipeline

**Code2Flow | AI Code-to-Flowchart Generator | [GitHub](#) • [View Live](#)**

*Python, Flask, JavaScript, HTML, CSS*

- Built an AI-powered tool that converts source code into deterministic ASCII-based flowcharts using Flask and Gemini API
- Designed a pipeline separating control-flow extraction from deterministic rendering for consistent output
- Established normalization and validation to handle LLM formatting variations and minimize API calls

## LEADERSHIP & ACTIVITIES

**Organizing Committee Member — Sports (Chess)**

*March 2025*

*Revels'25, MIT Manipal*

- Led logistics coordination for chess tournaments with 200+ participants, managing registration and bracket systems
- Collaborated with a 12-member cross-functional team ensuring 100% on-time event execution during cultural fest

## ACHIEVEMENTS

**LeetCode:** Solved 200+ problems across data structures and algorithms

**Achievers Scholarship:** Ranked in Top 5% of Information Technology cohort

**Academic Excellence:** Maintained 9.0+ GPA across all semesters at MIT Manipal

## ADDITIONAL

**Languages:** Hindi (Native), English (Proficient)

**Interests:** Competitive Programming, Chess, Badminton, Reading